Scaling to Huge Datasets & Online Learning

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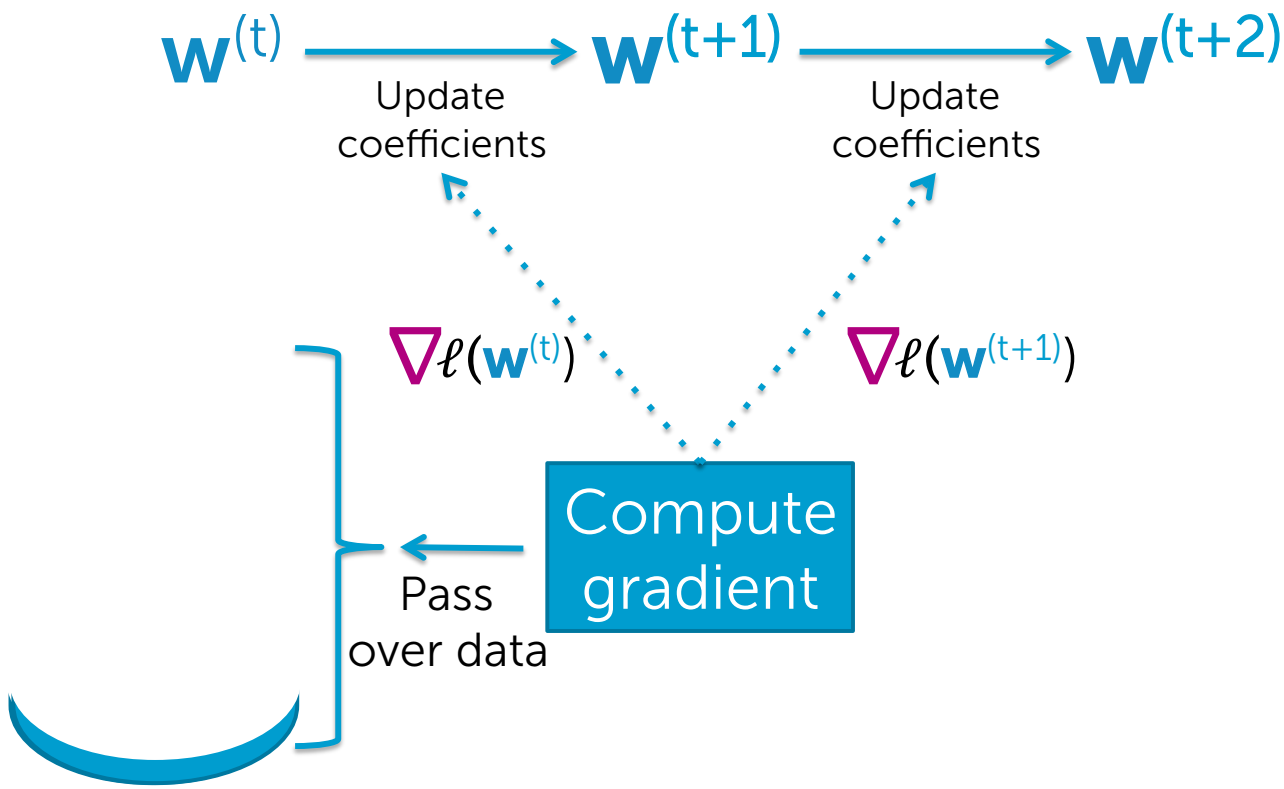
Why gradient ascent is slow…

Every update requires a full pass over data



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Data sets are getting huge, and we need them!

WWW
4.8B webpages

twitter
500M Tweets/day

Internet of Things
Sensors everywhere

YouTube
300 hours uploaded/min → 1 B users, ad revenue → 5B views/day

Need ML algorithm to learn from billions of video views every day, & to recommend ads within milliseconds

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ML improves (significantly) with bigger datasets

	1996	2006	2016
Data size	Small data	Big data	Bigger data
Model complexity	Complex • Needed for accuracy	Simple • Needed for speed	Complex • Needed for accuracy • Parallelism, GPUs, computer clusters
Hot topics	Kernels Graphical models	Logistic regression Matrix factorization	Boosted trees Tensor factorization Deep learning Massive graphical models

"The Unreasonable Effectiveness of Data"
[Halevy, Norvig, Pereira '09]

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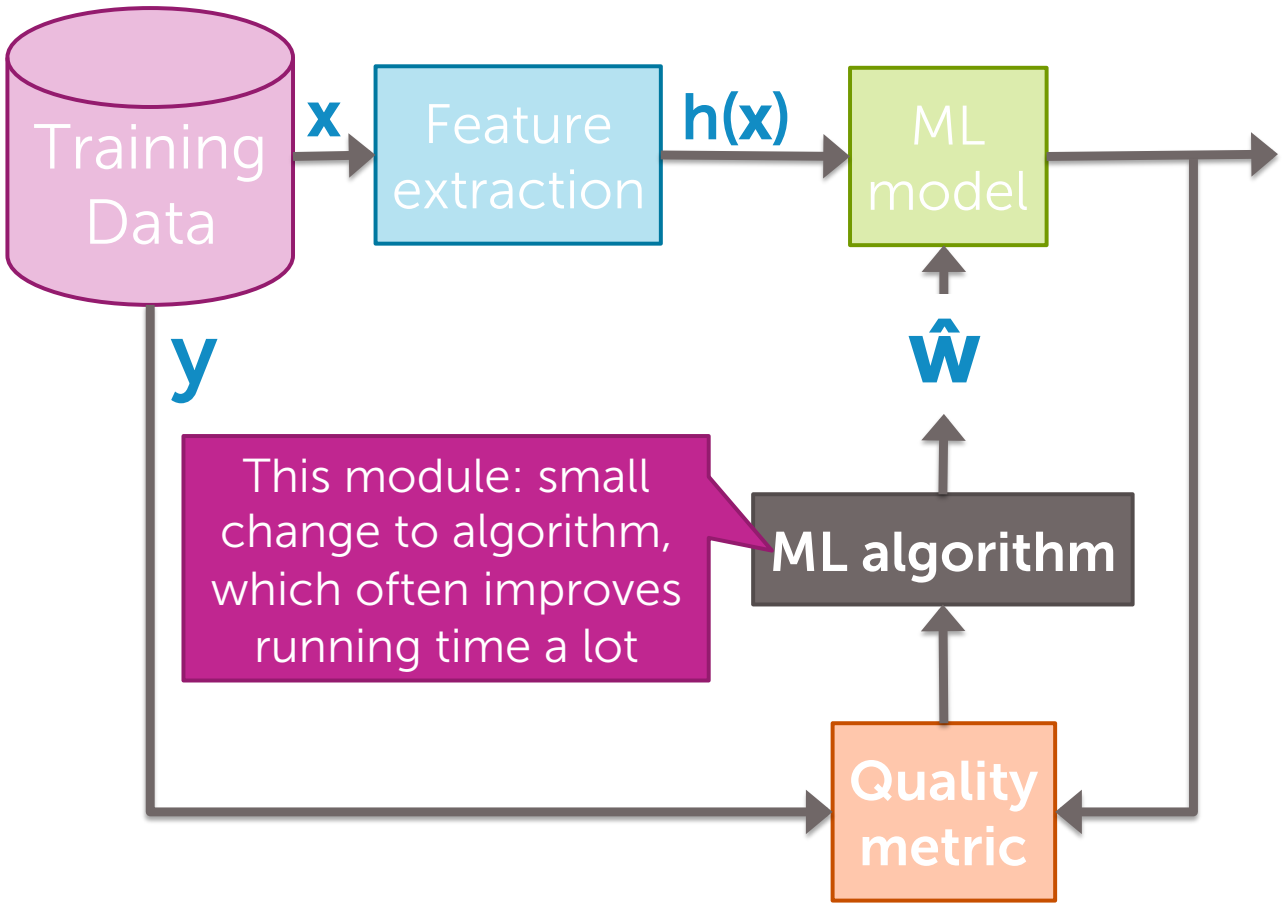

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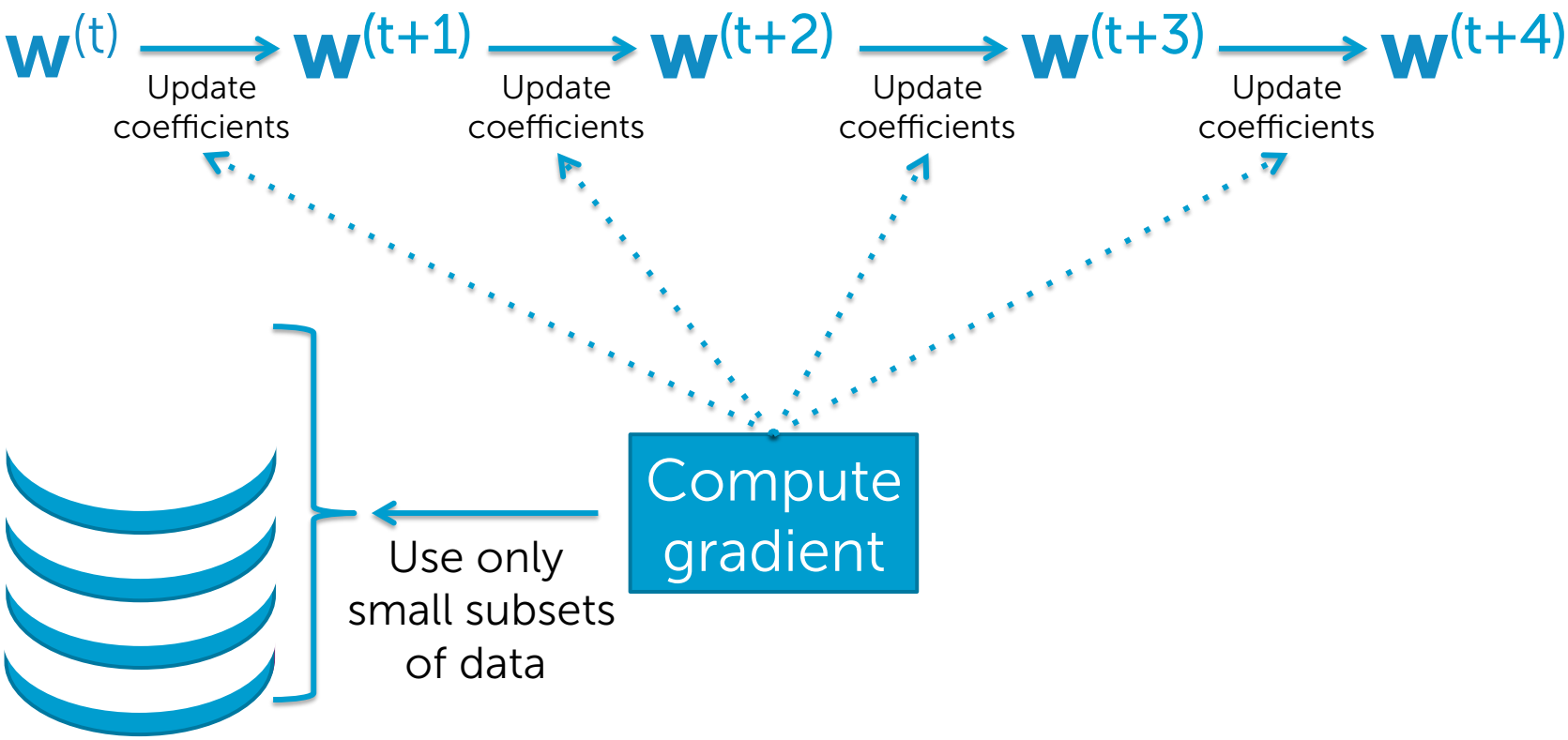
Stochastic gradient ascent

Many updates for each pass over data



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Learning, one data point at a time

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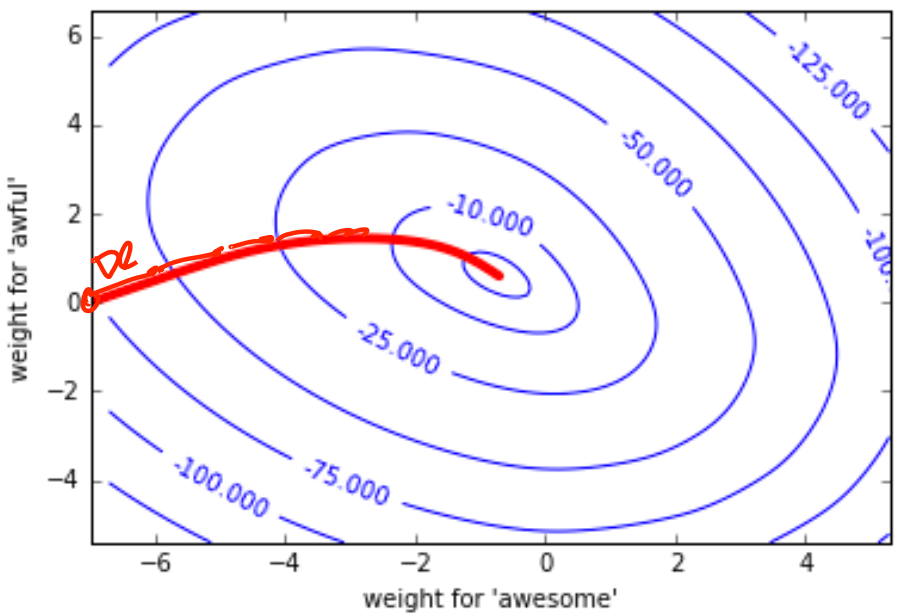
Gradient ascent



Algorithm:

while not converged

$$\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} + \eta \nabla \ell(\mathbf{w}^{(t)})$$

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How expensive is gradient ascent?

Sum over data points

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \underbrace{h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w})\Big)}_{\text{Contribution of data point } \mathbf{x}_i, y_i \text{ to gradient}}$$

$$\boxed{\frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}}$$

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Every step requires touching every data point!!!

Sum over data points

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

Time to compute contribution of $\mathbf{x}_i, y_i$	# of data points (N)	Total time to compute 1 step of gradient ascent
1 millisecond	1000	1 sec
1 second	1000	16.7 min
1 millisecond	10 million	2.8 hours
1 millisecond	10 billion	115.7 days

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Instead of all data points for gradient, use 1 data point only???

Sum over data points

Gradient ascent

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

Each time, pick different data point i

Stochastic gradient ascent

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} \approx \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

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Stochastic gradient ascent

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Stochastic gradient ascent for logistic regression

init $\mathbf{w}^{(1)}$=0, t=1

for i=1…N

until converged

for j=0,…,D

partial[j] = $\cancel{\sum_{i=1}^{N}} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\Big)$

$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta$ partial[j]

t ← t + 1

Sum over data points

Each time, pick different data point i

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Comparing computational time per step

Gradient ascent

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

Stochastic gradient ascent

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} \approx \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

Time to compute contribution of $\mathbf{x}_i, y_i$	# of data points (N)	Total time for 1 step of gradient	Total time for 1 step of stochastic gradient
1 millisecond	1000	1 second	1 millisecond
1 second	1000	16.7 minutes	1 sec
1 millisecond	10 million	2.8 hours	1 millisec
1 millisecond	10 billion	115.7 days	1 millisec

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Comparing gradient to stochastic gradient

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Which one is better??? Depends…

Algorithm	Time per iteration	Total time to convergence for large data		Sensitivity to parameters
		In theory	In practice	
Gradient	Slow for large data	Slower	Often slower	Moderate
Stochastic gradient	Always fast	Faster	Often faster	Very high

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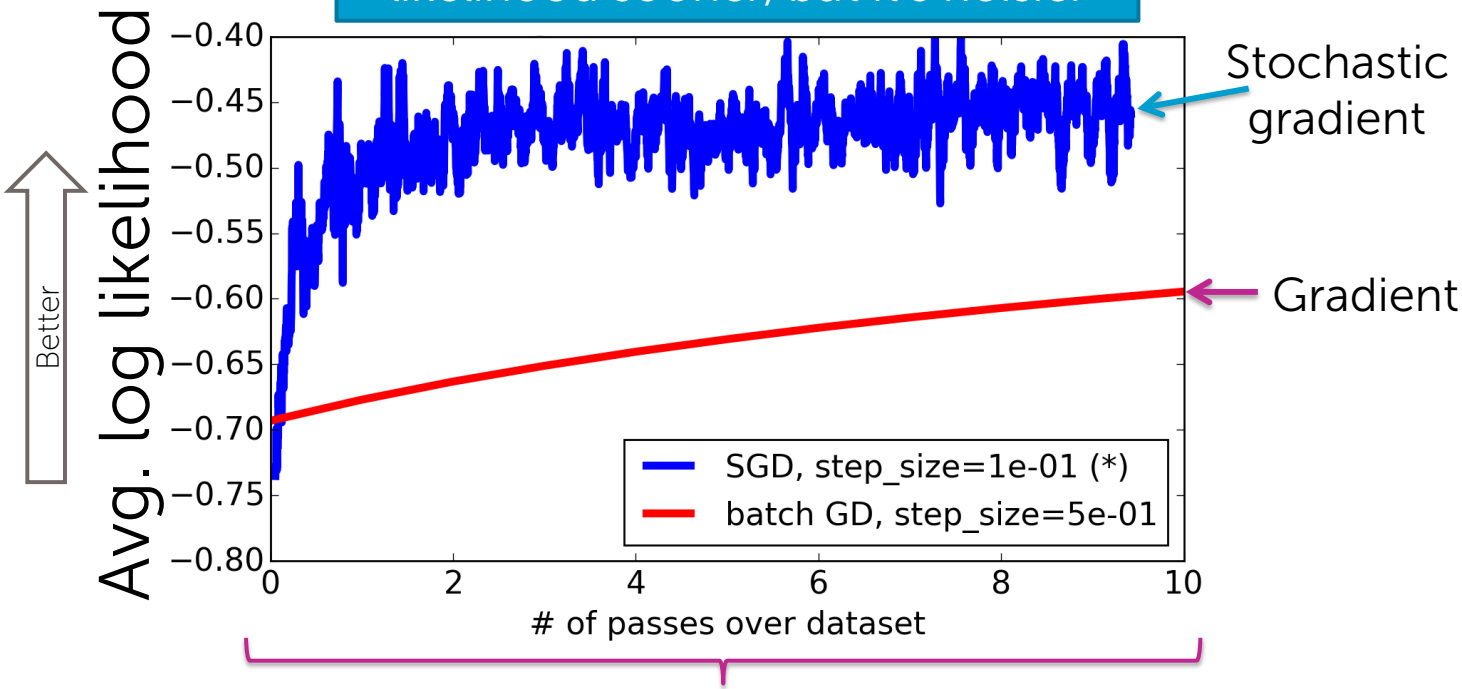
Comparing gradient to stochastic gradient

Stochastic gradient achieves higher likelihood sooner, but it's noisier



Total time proportional to number of passes over data

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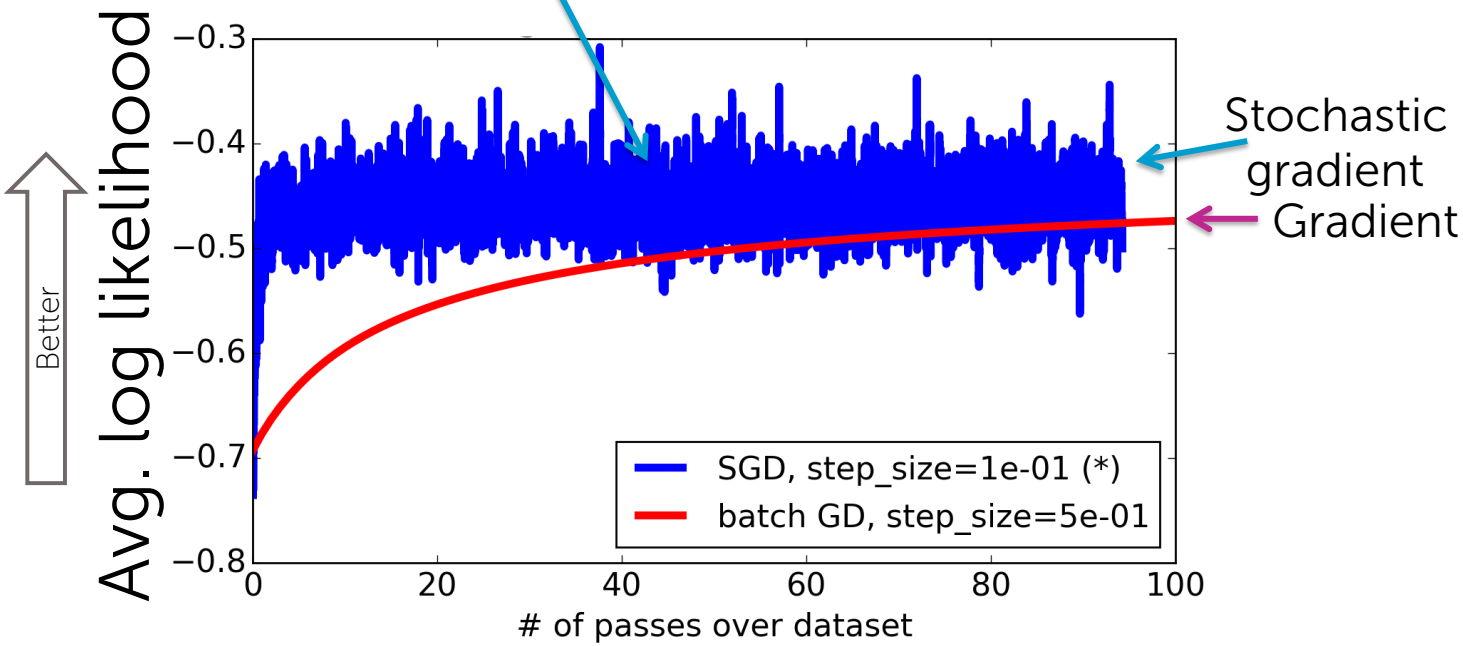
Eventually, gradient catches up

Note: should only trust "average" quality of stochastic gradient (more discussion later)



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Summary of stochastic gradient

- Tiny change to gradient ascent
- Much better scalability
- Huge impact in real-world
- Very tricky to get right in practice

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Why would stochastic gradient ever work???

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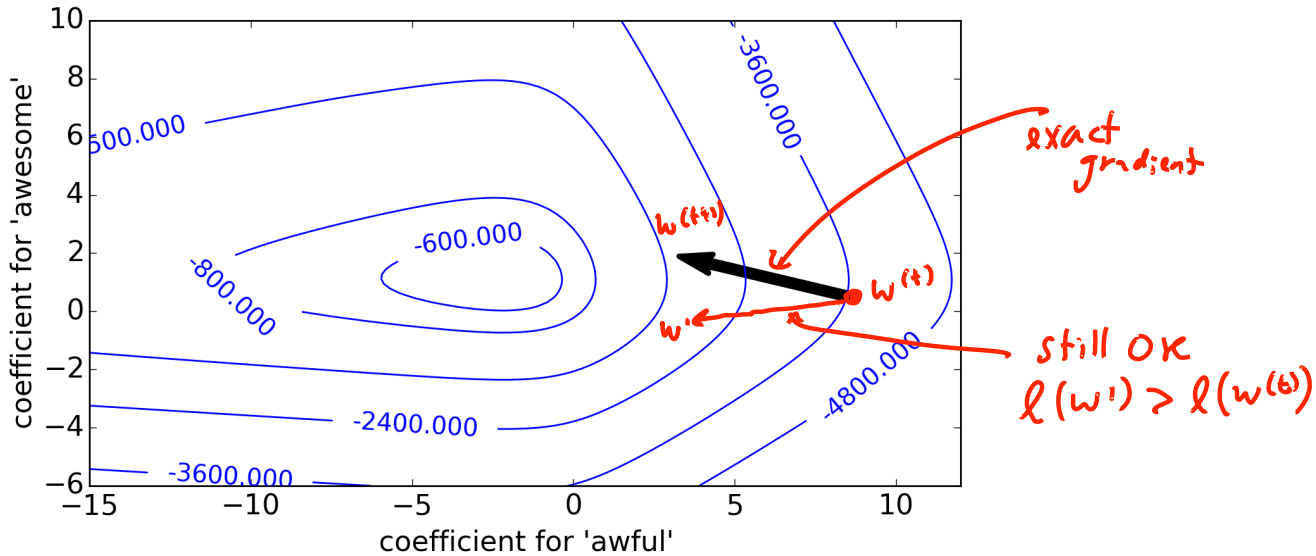
Gradient is direction of steepest ascent



Gradient is "best" direction, but
any direction that goes "up" would be useful

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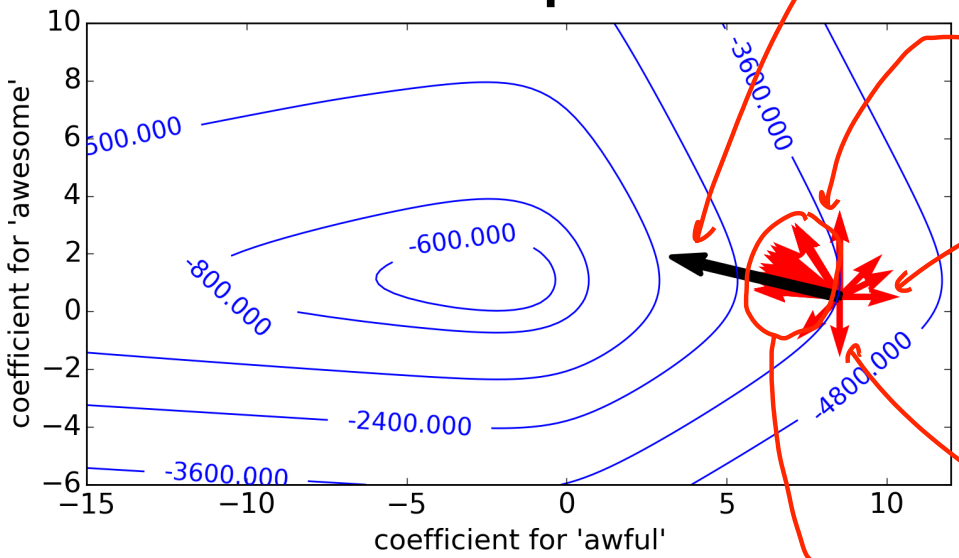
In ML, steepest direction is sum of "little directions" from each data point



gradient

bad directions

Sum over data points

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \underbrace{\frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}}_{\text{contribution from each } x_i y_i}$$

For most data points, contribution points "up"

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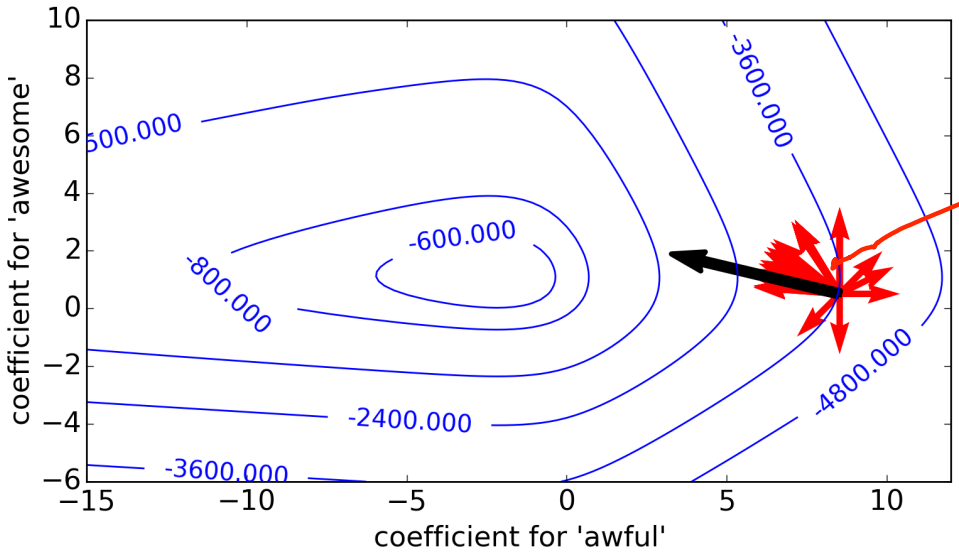
Stochastic gradient: pick a data point and move in direction



pick one of these

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} \approx \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$$

Most of the time, total likelihood will increase

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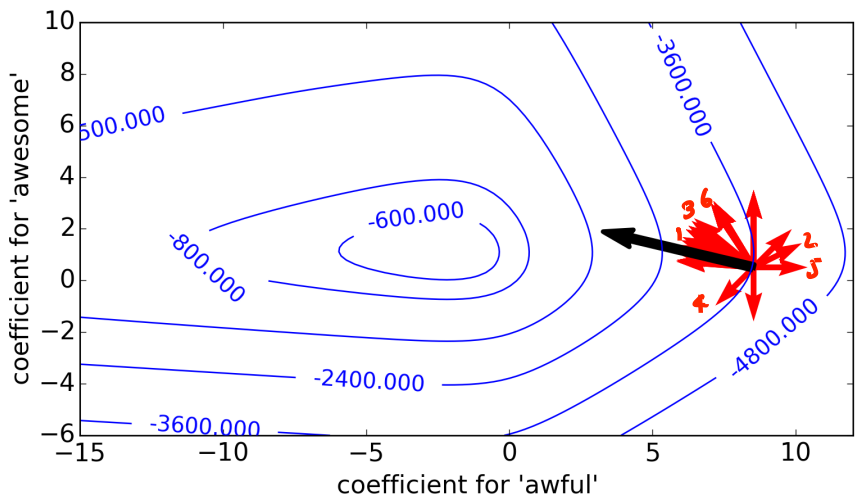
Stochastic gradient ascent:

Most iterations increase likelihood,
but sometimes decrease it ➔
On average, make progress



until converged

  for i=1,…,N

    for j=0,…,D

    $w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \dfrac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$

    $t \leftarrow t + 1$

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Convergence path

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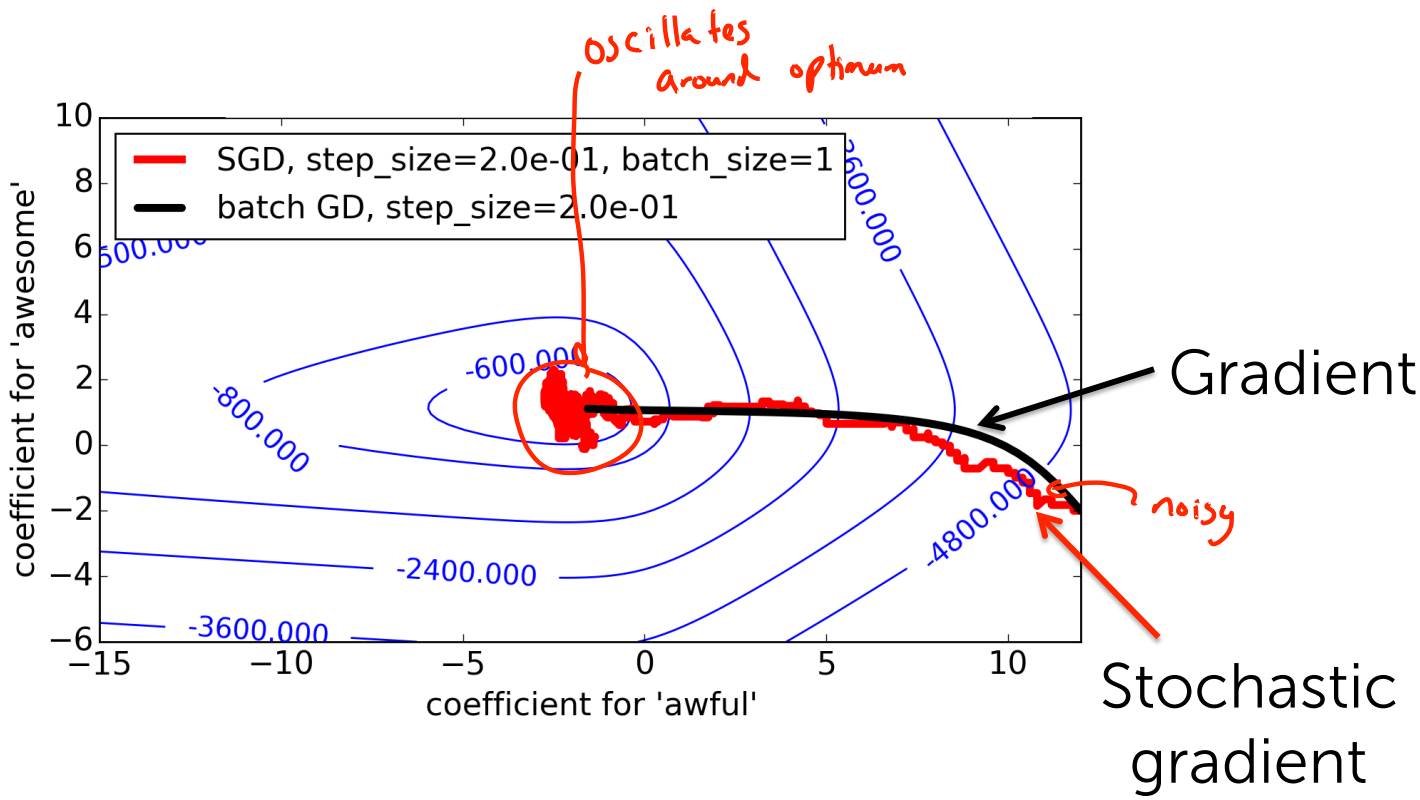
Convergence paths



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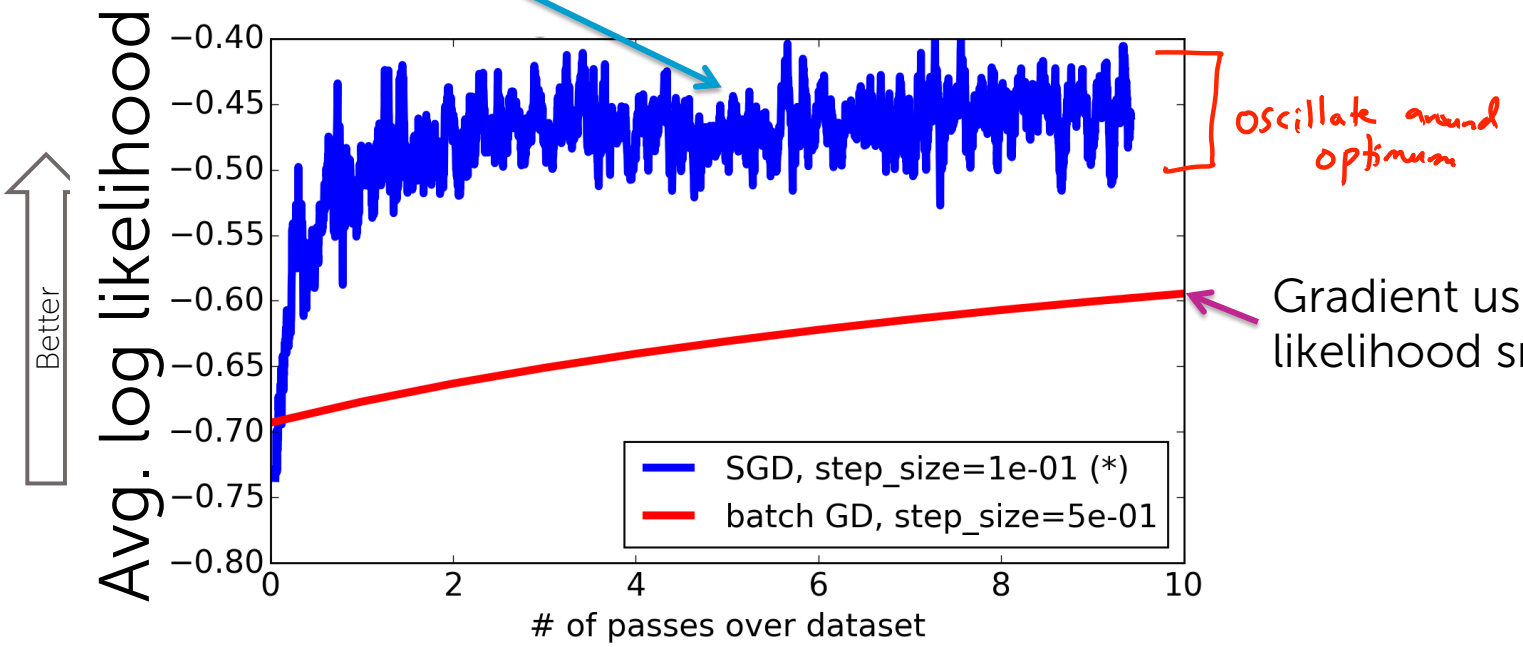
Stochastic gradient convergence is "noisy"

Stochastic gradient makes "noisy" progress



Gradient usually increases likelihood smoothly

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Summary of why stochastic gradient works

Gradient finds direction of steeps ascent

Gradient is sum of contributions from each data point

Stochastic gradient uses direction from 1 data point

On average increases likelihood, sometimes decreases

Stochastic gradient has "noisy" convergence

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Stochastic gradient:
practical tricks

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Stochastic gradient ascent

init $\mathbf{w}^{(1)}=0$, t=1

until converged

  for i=1,…,N

    for j=0,…,D

      $w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \dfrac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$

    $t \leftarrow t + 1$

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Order of data can introduce bias

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1
2	1	+1
4	1	+1
1	1	+1
2	1	+1

Stochastic gradient updates parameters 1 data point at a time

Systematic order in data can introduce significant bias, e.g., all negative points first, or temporal order, younger first, or …

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Shuffle data before running stochastic gradient!

x[1] = #awesome	**x**[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1
2	1	+1
4	1	+1
1	1	+1
2	1	+1

Shuffle rows

x[1] = #awesome	**x**[2] = #awful	y = sentiment
1	1	+1
3	3	-1
0	2	-1
4	1	+1
2	1	+1
2	4	-1
0	1	-1
0	3	-1
2	1	+1

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Stochastic gradient ascent

Before running stochastic gradient, make sure data is shuffled

Shuffle data

init $\mathbf{w}^{(1)}=0$, t=1

until converged

  for i=1,…,N

    for j=0,…,D

      $w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \dfrac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j}$

    $t \leftarrow t + 1$

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Choosing the step size η

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Picking step size
for **stochastic gradient**
is very similar to
picking step size
for **gradient**

But stochastic gradient
is a lot more unstable… ☹

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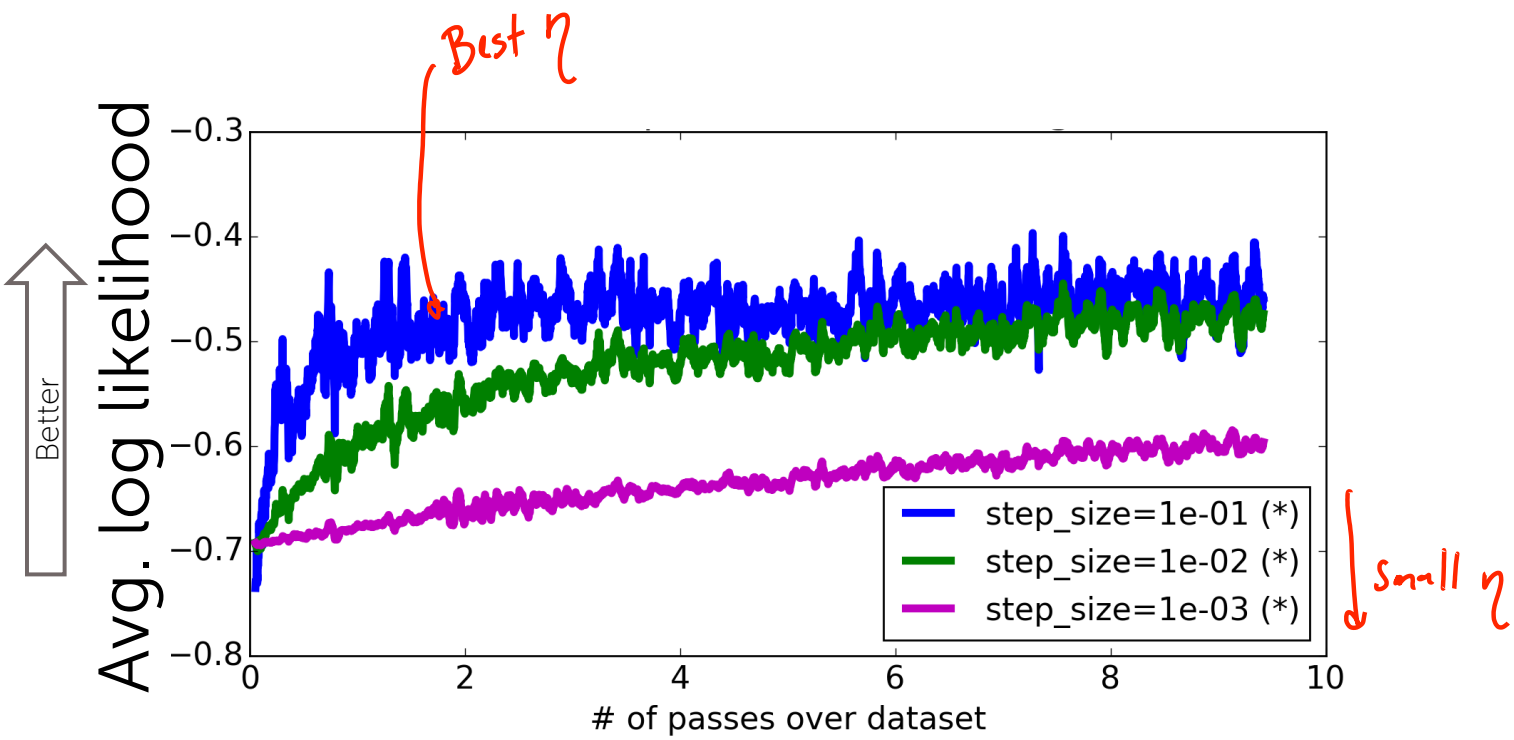
If step size is too small, stochastic gradient slow to converge



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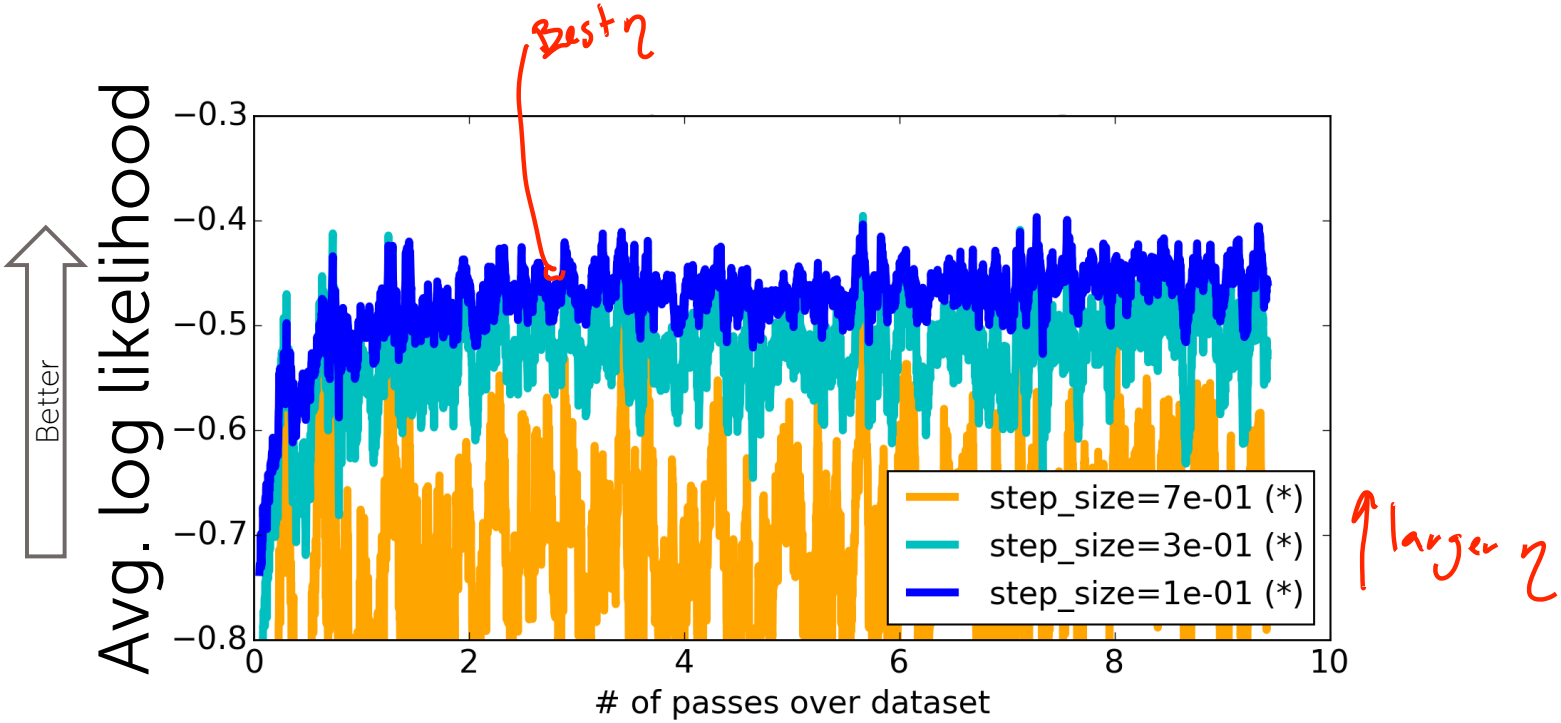
If step size is too large, stochastic gradient oscillates



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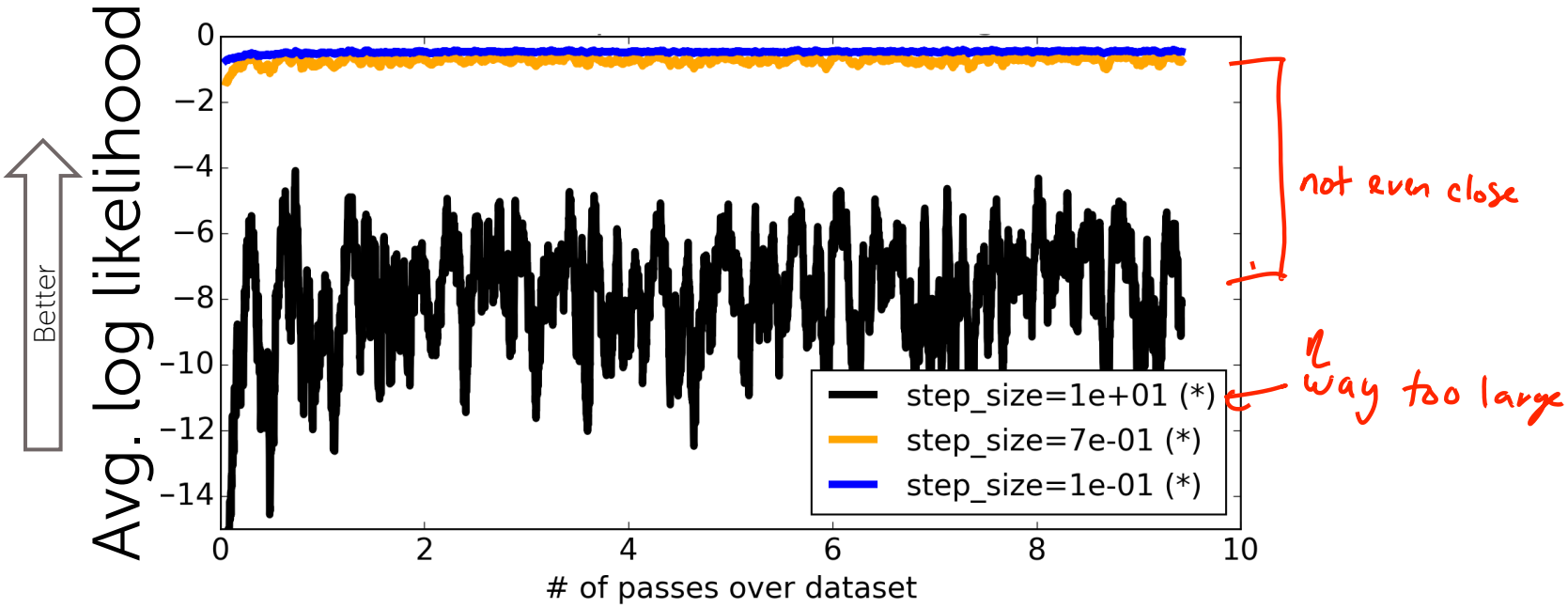
If step size is very large, stochastic gradient goes crazy ☹



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Simple rule of thumb for picking step size η similar to gradient

- Unfortunately, picking step size requires a lot a lot of trial and error, much worst than gradient ☹
- Try a several values, exponentially spaced
  - **Goal**: plot learning curves to
    - find one η that is too small
    - find one η that is too large

- *Advanced tip*: step size that decreases with iterations is very important for stochastic gradient, e.g., $\eta_t = \frac{\eta_0}{t}$ ← constant ← iteration #

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Don’t trust the last coefficients… ☹

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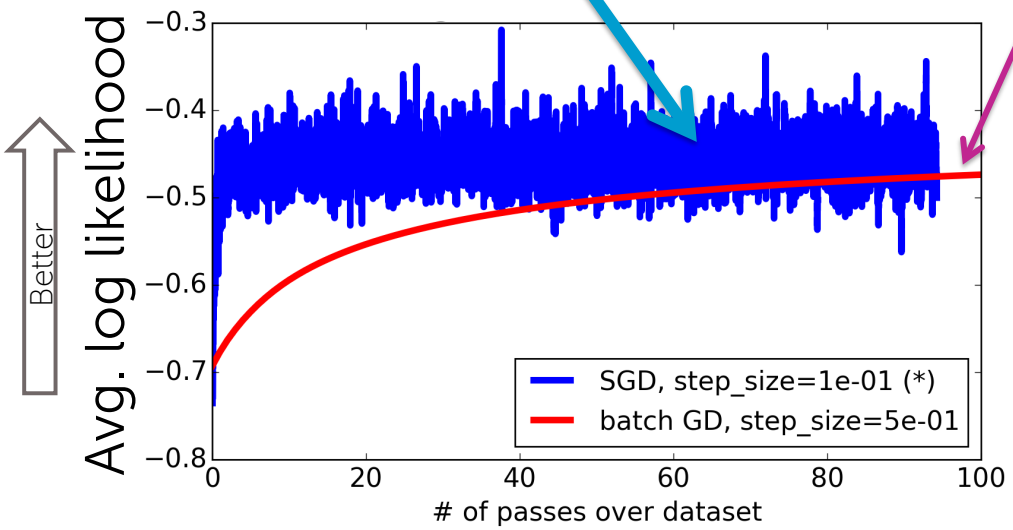
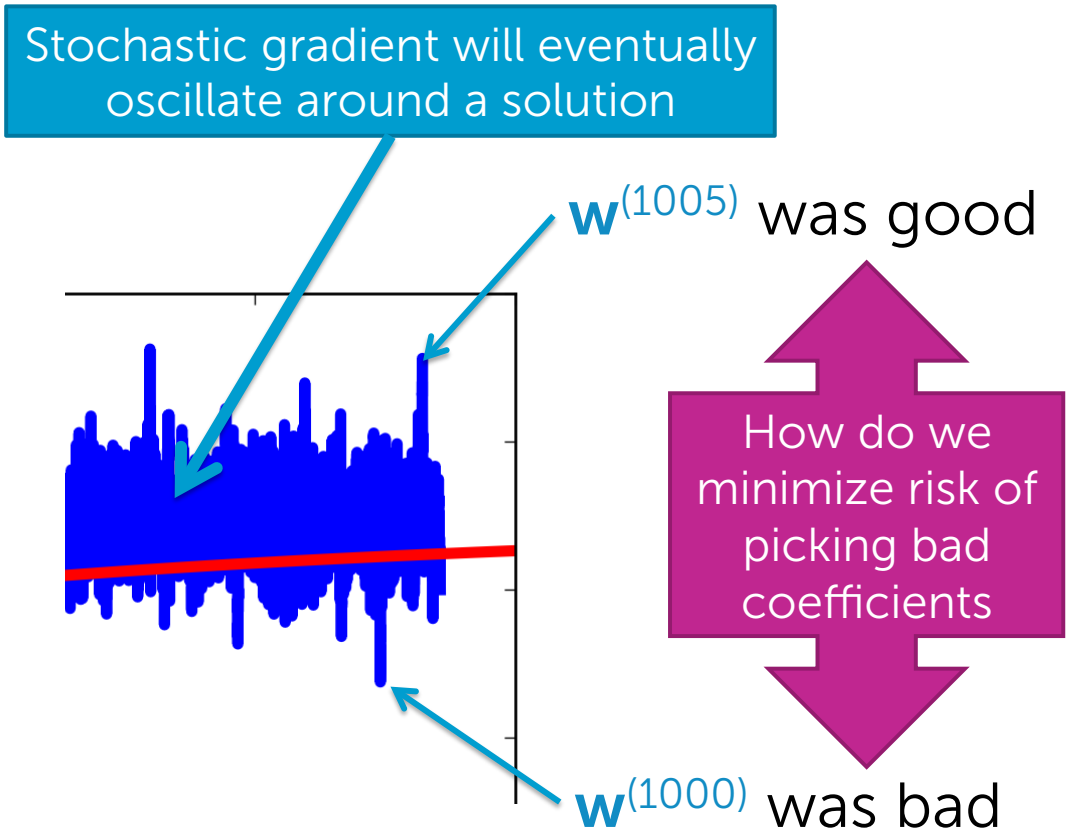
Stochastic gradient never fully "converges"

Stochastic gradient will eventually oscillate around a solution

Gradient will eventually stabilize on a solution



The last coefficients may be really good or really bad!! ☹



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Stochastic gradient returns average coefficients

- Minimize noise:
  don't return last learned coefficients

- Instead, output average:

$$\hat{\mathbf{w}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{w}^{(t)}$$

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Learning from batches of data

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Gradient/stochastic gradient: two extremes

Stochastic gradient:
1 point/iteration

Gradient:
N points/iteration

Mini-batches:
B points/iteration

Reduces noise,
increases stability

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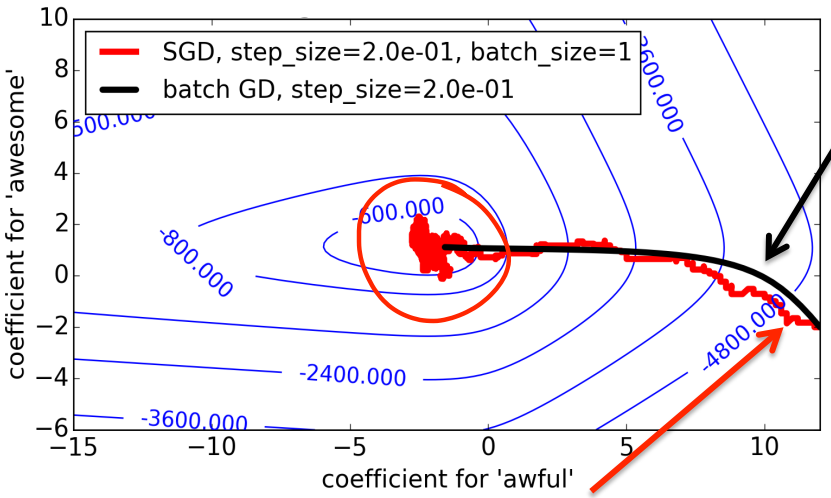
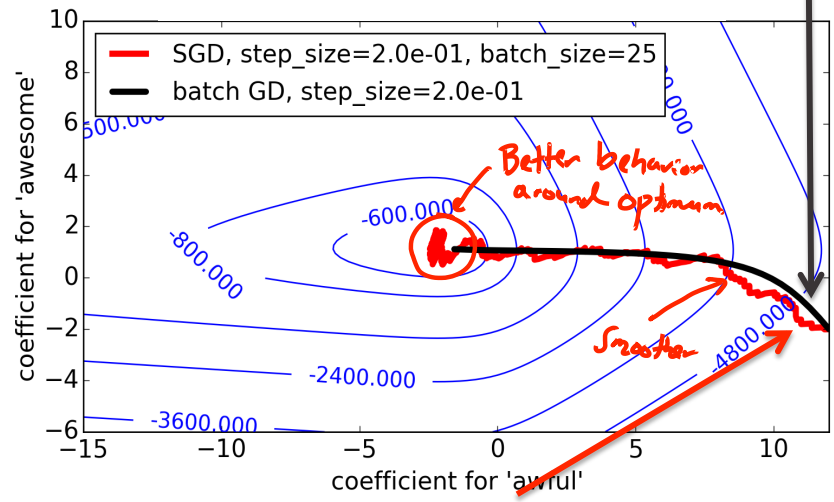
Convergence paths

Gradient



Stochastic gradient
Batch size = 1



Stochastic gradient
Batch size = 25

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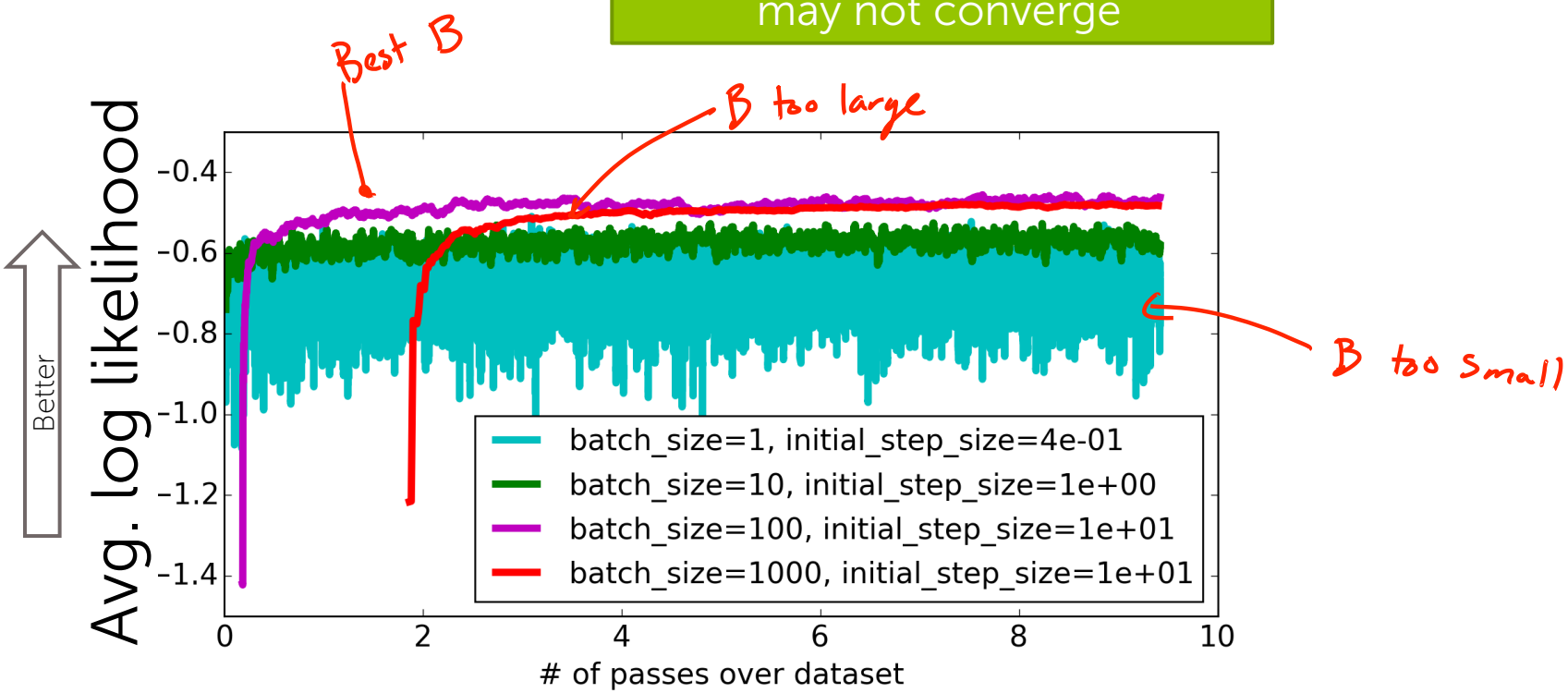
Batch size effect

Too large → Slow convergence

Too small → Noisy, may not converge



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Stochastic gradient ascent with mini-batches

Shuffle data

init $\mathbf{w}^{(1)}=0$, t=1

until converged

for k=0,…,N/B-1

for j=0,…,D

$$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \sum_{i=1+kB}^{(k+1)*B} \frac{\partial \ell_i(\mathbf{w})}{\partial w_j}$$

t ← t + 1

For each mini-batch

Sum over data points in mini-batch k

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Measuring convergence

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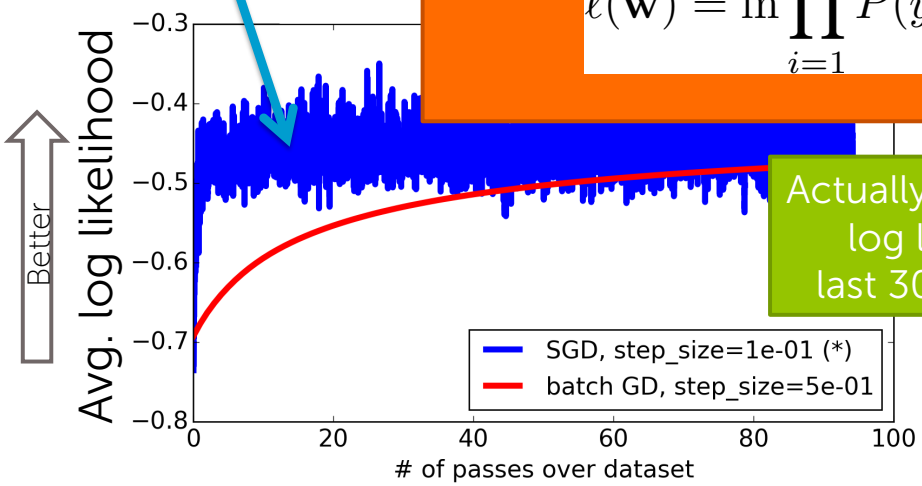
How did we make these plots???

Need to compute log likelihood of data at every iteration???

➔ Really really slow, product over all data points!

$$\ell(\mathbf{w}) = \ln \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$



Actually, plotting average log likelihood over last 30 mini-batches...

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Computing log-likelihood during run of stochastic gradient ascent

init $\mathbf{w}^{(1)}$=0, t=1

until converged

for i=1,…,N

for j=0,…,D

partial[j] = $h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\Big)$

$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta$ partial[j]

t ← t + 1

Log-likelihood of data point i is simply:

$$\ell_i(\mathbf{w}^{(t)}) = \begin{cases} \ln P(y = +1 \mid \mathbf{x}_i, \mathbf{w}^{(t)}), & \text{if } y_i = +1 \\ \ln\big(1 - P(y = +1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\big), & \text{if } y_i = -1 \end{cases}$$

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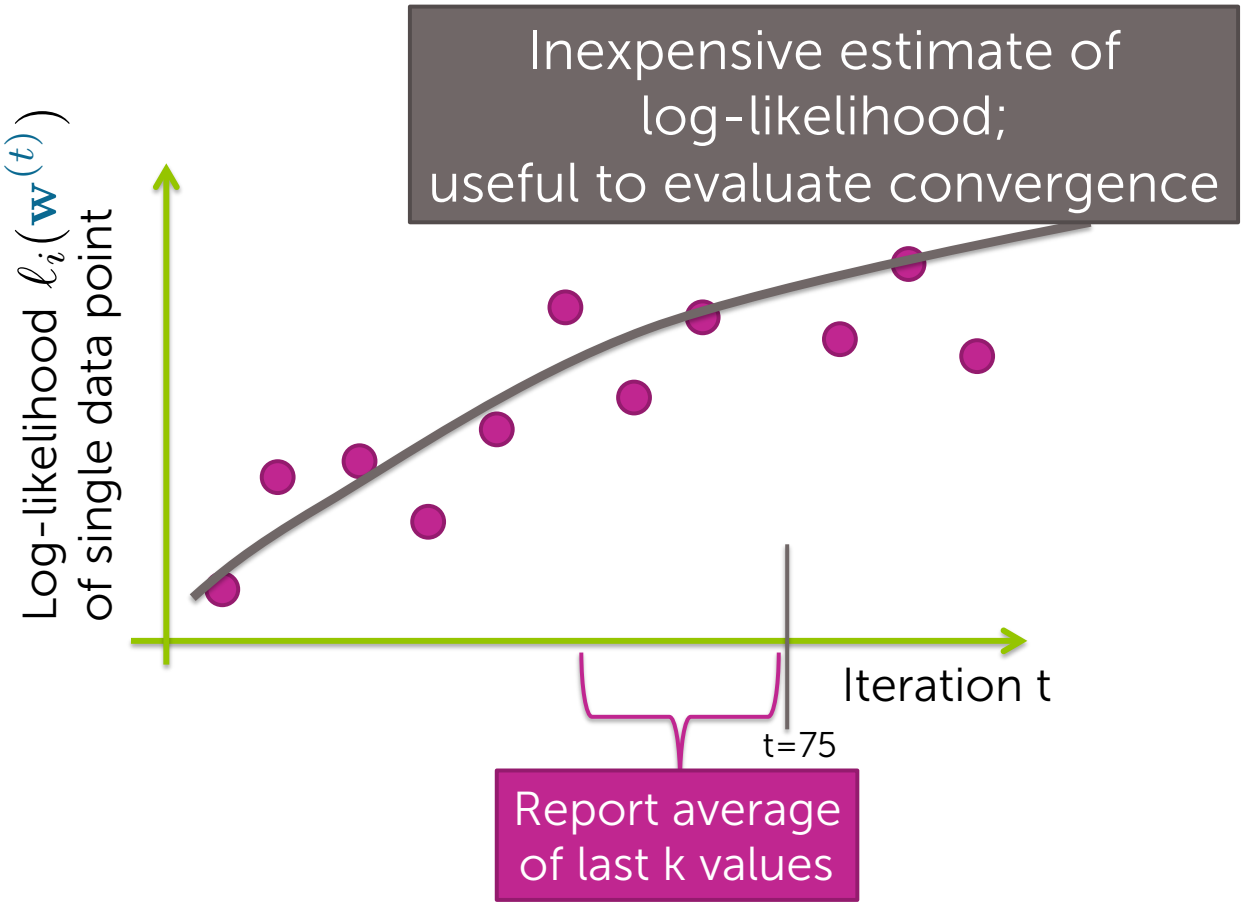
Estimate log-likelihood with sliding window



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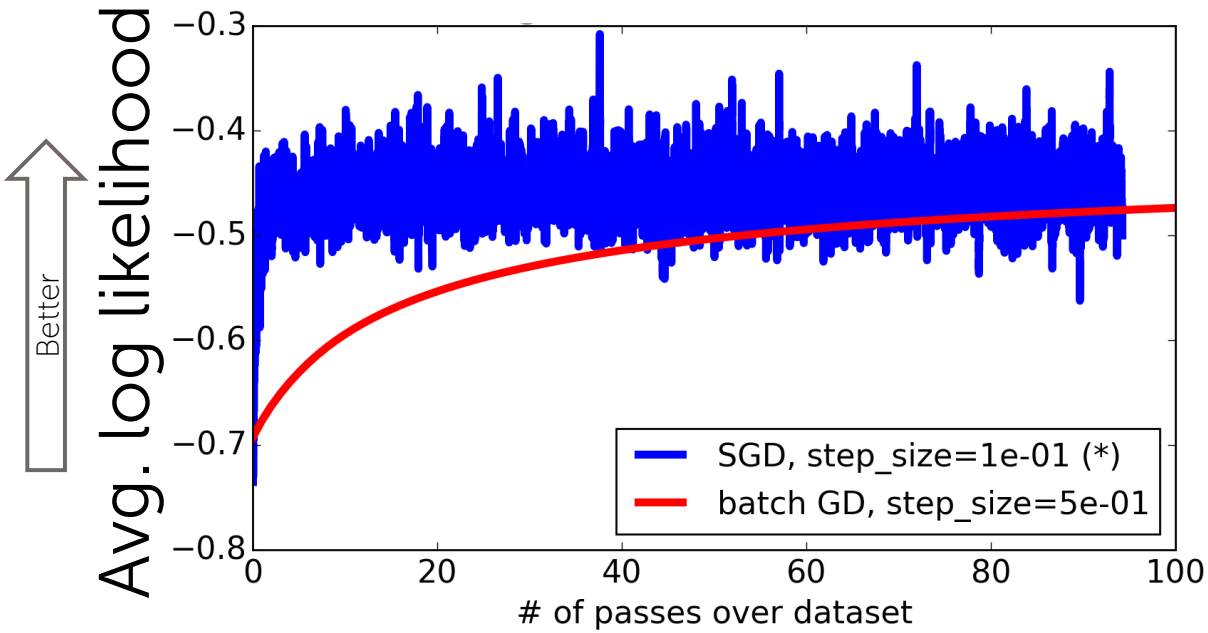
That's what average log-likelihood meant... ☺

(In this case, over last k=30 mini-batches, with batch-size B = 100)



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Adding regularization

OPTIONAL

Consider specific total cost

$\max_{\mathbf{w}}$

Total quality =

measure of fit - measure of magnitude of coefficients

$$\underbrace{\ell(\mathbf{w})}_{\text{log data likelihood}} \qquad \underbrace{||\mathbf{w}||_2^2}_{L_2 \text{ penalty}}$$

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Gradient of L_2 regularized log-likelihood

Total quality =

measure of fit - measure of magnitude of coefficients

$$\underbrace{\ell(\mathbf{w})}_{\text{measure of fit}} \quad \underbrace{\lambda \, ||\mathbf{w}||_2^2}_{\text{measure of magnitude of coefficients}}$$

$$\text{Total derivative} = \sum_{i=1}^{N} \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j} - 2\lambda \, w_j$$

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Stochastic gradient for regularized objective

$$\text{Total derivative} = \sum_{i=1}^{N} \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j} - 2\lambda w_j$$

- What about regularization term?

Each time, pick different data point i

Stochastic gradient ascent

$$\text{Total derivative} \approx \frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j} - \frac{2}{N}\lambda w_j$$

Each data point contributes 1/N to regularization

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Stochastic gradient ascent with regularization

Shuffle data

init $\mathbf{w}^{(1)}=0$, t=1

until converged

for i=1,…,N

for j=0,…,D

$$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \left(\frac{\partial \ell_i(\mathbf{w})}{\partial \mathbf{w}_j} - \frac{2}{N} \lambda \, w_j \right)$$

t ← t + 1

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Online learning: Fitting models from streaming data

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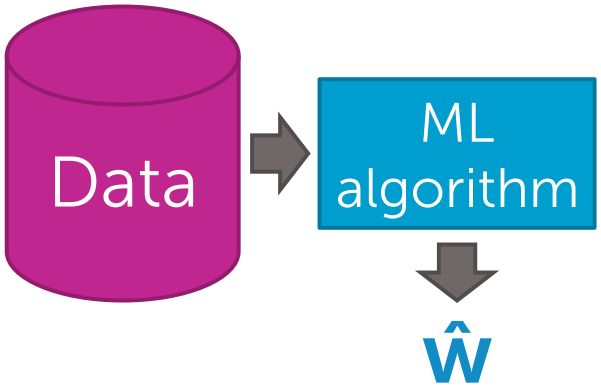
Batch vs online learning

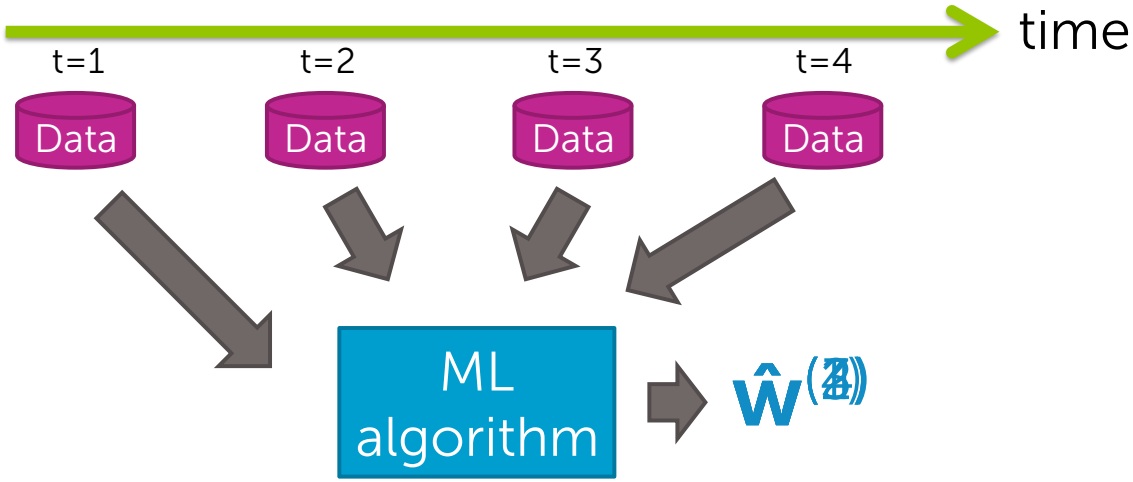
Batch learning

- All data is available at start of training time



Online learning

- Data arrives (streams in) over time
  - Must train model as data arrives!



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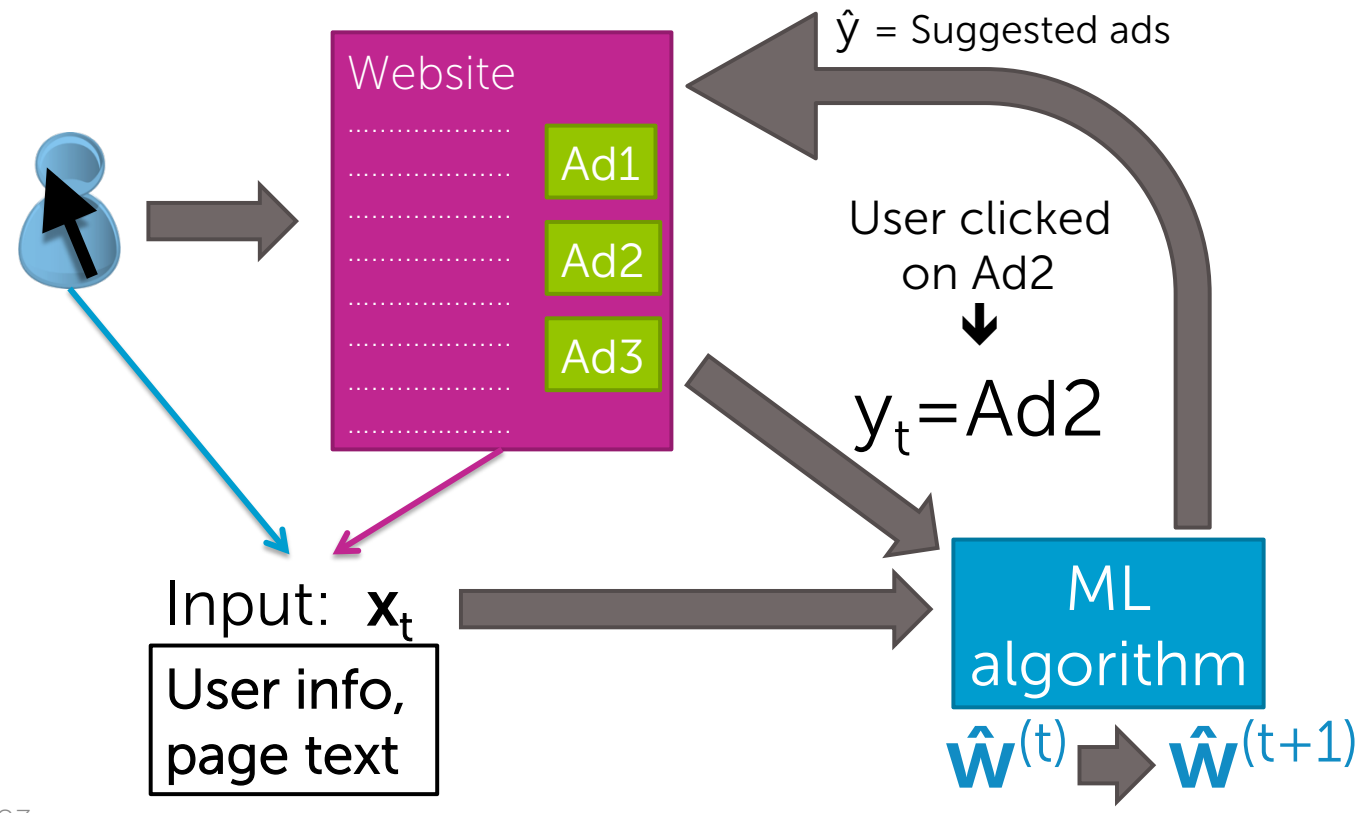
Online learning example:
Ad targeting



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Online learning problem

- Data arrives over each time step t:
  - **Observe input x_t**
    - Info of user, text of webpage
  - **Make a prediction $\hat{y}_t$**
    - Which ad to show
  - **Observe true output y_t**
    - Which ad user clicked on

Need ML algorithm to update coefficients each time step!

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Stochastic gradient ascent can be used for online learning!!!

- init $\mathbf{w}^{(1)}=0$, t=1
- Each time step t:
  - Observe input x_t
  - Make a prediction $\hat{y}_t$
  - Observe true output y_t
  - Update coefficients:

for j=0,…,D

$$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta \frac{\partial \ell_t(\mathbf{w})}{\partial \mathbf{w}_j}$$

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Summary of online learning

Data arrives over time

Must make a prediction every time new data point arrives

Observe true class after prediction made

Want to update parameters immediately

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Updating coefficients immediately: Pros and Cons

Pros

- Model always up to date ➔ Often more accurate
- Lower computational cost
- Don't need to store all data, but often do anyway

Cons

- Overall system is *much* more complex
  - Bad real-world cost in terms of $$$ to build & maintain

Most companies opt for systems that save data and update coefficients every night, or hour, week,…

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Summary of scaling to huge datasets & online learning

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Scaling through parallelism

Multicore processors

Computer clusters

Huge opportunity for scaling

Require parallel & distributed ML algorithms

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What you can do now…

- Significantly speedup learning algorithm using stochastic gradient
- Describe intuition behind why stochastic gradient works
- Apply stochastic gradient in practice
- Describe online learning problems
- Relate stochastic gradient to online learning

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